

**Esercizio 1.** Let  $R$  pp formula and let  $M, N$  ec models of  $\mathcal{T}^\pm$ . Suppose that  $R$  has the same number of solutions in  $M, N$ .

*Proof.* Suppose  $M$  has exactly  $k$  solutions. We can express "there are at most  $k$  solutions" as:

$$\forall x_1, \dots, x_k, x_{k+1} \left( \bigwedge_{i=1, \dots, k+1} R(x_i) \rightarrow \bigvee_{i < j} x_i = x_j \right)$$

This is preserved by 2) of the lemma, as "=" will be expressible by a pp-type. We can express the negation of "there are at least  $k$  solutions" as:

$$\forall x_1, \dots, x_k \left( \bigwedge_{i=1, \dots, k} R(x_i) \rightarrow \bigvee_{i < j} x_i = x_j \right)$$

This is preserved by 2) of the lemma as "=" will be expressible by a pp-type.  $\square$